

AJAY KUMAR AVULA

Göteborg, Sweden • (+46) 793456499 • ajaykumaravula999@gmail.com
linkedin.com/in/ajay-kumar-avula/ • github.com/ajaykumaravula

PROFESSIONAL SUMMARY

Results-driven MLOps and Cloud DevOps Engineer with 3+ years of experience automating infrastructure, optimizing CI/CD pipelines, and architecting production-grade AI microservices. Armed with a Master's in AI and Automation, specializing in bridging the gap between data science and cloud infrastructure by containerizing ML pipelines, managing model serialization, and orchestrating high-availability deployments. Certified AWS Solutions Architect Associate with a proven track record of reducing infrastructure costs and eliminating environmental drift across distributed systems. Actively learning Swedish to enhance cross-cultural collaboration and global teamwork.

PROFESSIONAL EXPERIENCE

Gaddr — Stockholm, Sweden | AI Fullstack Developer Intern

Feb 2025 – Present

- **AI Microservice Architecture:** Architected and deployed an end-to-end, high-performance predictive analytics web application utilizing a stateless Flask REST API to deliver real-time model inference.
- **Model Comparison Pipeline:** Engineered an automated training and evaluation pipeline using Scikit-Learn to benchmark tree-based ensemble and gradient boosting frameworks (Random Forest, XGBoost, LightGBM, CatBoost), evaluating performance via MAE, RMSE, and R^2 scores to capture over 80% of market variance.
- **Model Serialization:** Streamlined production serving infrastructure by implementing model serialization (joblib) to pre-load binary artifacts into runtime memory upon server startup, drastically minimizing API inference latency.
- **Docker Containerization:** Containerized the entire application ecosystem utilizing Docker, authoring multi-stage Dockerfile configurations with lightweight base images (python:3.10-slim) to reduce image footprints and guarantee deterministic builds.
- **Production Hardening:** Designed a robust production runtime by resolving detached container stream termination defects (debug=False) and configuring explicit host-to-container port mapping (-p 5001:5001) for secure endpoint exposure.
- **Orchestration & GitOps:** Deployed and managed distributed Java, Python, and TypeScript microservices on AWS EKS, creating reusable Helm charts for streamlined, environment-agnostic deployments while spearheading the transition to GitOps-based deployment models using Argo CD.
- **Infrastructure as Code:** Automated provisioning and lifecycle management of Kubernetes clusters using Terraform, ensuring reproducible infrastructure and reducing configuration drift.
- **Cloud Cost Optimization:** Reduced infrastructure costs by 30% by migrating workloads to AWS EKS, optimizing resource utilization, and implementing right-sizing strategies for compute workloads.
- **CI/CD & Observability:** Developed end-to-end GitLab CI/CD pipelines for automated build, test, and deployment, while implementing centralized observability using Prometheus, Loki, and Grafana for real-time proactive alerting.

I CUBE SOLUTIONS — Hyderabad, India | Associate DevOps Engineer

Mar 2022 – Aug 2024

- **Cloud Infrastructure:** Automated the provisioning and management of core AWS services—including IAM, EC2, VPC, S3, RDS, and Route 53—using Terraform for scalable and consistent cloud infrastructure.
- **Automation & Configuration:** Developed and improved automation workflows for infrastructure and application deployments using Terraform, Kubernetes, and Helm, contributing to more reliable and scalable systems.
- **CI/CD Pipeline Optimization:** Maintained and optimized CI/CD pipelines with GitLab CI, integrating automated code quality testing (SonarQube), security scanning, and container monitoring tools (Grafana, Prometheus).
- **Team Mentorship:** Provided technical support and guidance to new team members, fostering a collaborative knowledge-sharing environment through documentation and hands-on training with Kubernetes and GitLab CI.

I CUBE SOLUTIONS — Hyderabad, India | DevOps Engineer Intern

Mar 2022 – Promotion

- Assisted in troubleshooting build and deployment issues, learning industry best practices in cloud operations.
- Supported the core DevOps team in automating infrastructure provisioning and CI/CD pipeline tasks.
- Gained hands-on experience with cloud platforms and containerization, contributing to routine deployments and monitoring activities.

EDUCATION

University West — Trollhättan, Sweden

Master's Degree in AI and Automation

Blekinge Institute of Technology — Karlskrona, Sweden

Master's Degree in Software Engineering

Jawaharlal Nehru Technological University — India

Bachelor's Degree in Computer Science Engineering

TECHNICAL STRENGTHS

Programming Languages Python, Bash, Java, JavaScript, PowerShell

MLOps & AI Frameworks Scikit-Learn, XGBoost, LightGBM, CatBoost, Model Serialization (joblib/pickle), Regression Analysis, Ensemble Learning, Pandas, NumPy

Container & Orchestration Docker, Kubernetes, Helm, AWS EKS, Environment Isolation, Port Mapping

IaC & GitOps Terraform, Ansible, Argo CD, Declarative Infrastructure

CI/CD & Automation GitLab CI, Jenkins, Automation Scripting

Monitoring & Observability Grafana, Prometheus, ELK Stack (Elasticsearch, Logstash, Kibana), Loki, Mimir

Cloud, DB & Protocols AWS, Azure, MySQL, Kafka, TCP/IP, DNS, SNMP, DHCP, FTP

CERTIFICATIONS & ACHIEVEMENTS

- Amazon Web Services Certified Solutions Architect – Associate (AWS-SAA)
- Amazon Web Services Certified AI Practitioner & Certified Cloud Practitioner
- IBM Certified Docker Essentials & IBM Certified Kubernetes Service
- Student Ambassador for International Exchange Students at BTH, Sweden